Can AI models predict customer churn early enough in telecommunications businesses to increase a company's revenue?

Introduction

Customer retention is one of the biggest challenges faced in the telecommunications industry due to high competition and the cost of acquiring new customers. With the growth of artificial intelligence and machine learning, companies are starting to use predictive models to identify customers who are likely to discontinue their services, also known as customer churn. Previous research shows that machine learning algorithms can accurately identify churn using customer data such as user behavior, personal information, and interaction with the product. However, the ability of these models to have financial value depends on how effectively companies use the predictions to improve their strategies. This research explores whether AI models can predict customer churn early enough for telecommunications businesses to take preventive actions and increase revenue.

Telecommunications companies operate in a highly competitive sector of the market where keeping existing customers is super important for maintaining revenue and profit growth. Customer churn, which occurs when customers stop using a company's services, creates major financial losses because businesses must spend additional resources and money to acquire new customers. As a result, telecommunications companies have increasingly focused on predicting customer behavior before their customers decide to leave. Traditional methods of identifying customer churn usually rely on historical claims or customer complaints, which may happen too late for companies to prevent customer loss. Artificial intelligence and machine learning provide a potential solution by analyzing large amounts of customer data and identifying patterns that indicate a higher likelihood of churn.

Methodology

A. Research Design

This research uses quantitative machine learning methodology to evaluate whether artificial intelligence models can accurately predict customer churn. This research follows a standard classification framework, where historical data is used to train predictive models that classify customers into one of two categories: Churn(customer leaves the company), or Non-Churn(customer remains in the company).

The goal is not only to maximize prediction accuracy but also determine whether predictions can be early enough to give enough insight into what the business might need to fix or change.

B. Dataset

The main dataset used in this study was the Customer Churn Dataset published by Dimitar Tokmakov (2024), which contains specific customer records from a leading Bulgarian telecommunications operator. Unlike existing publicly available datasets that focus on residential customers, this dataset contains business customers ranging from small business to large enterprises. The dataset includes the following:

- Personal identification number of the business customer
- Value of a customer to the telecom company
- ZIP code of the company
- The name of their Key Account Manager
- The number of active subscribers
- The number of non-active subscribers
- The number of suspended subscribers in last 6 months
- Total subscribers
- Average subscriber revenue from mobile services
- Average subscriber revenue for fixed services
- Total Revenue
- Average revenue per user

The target variable, Churn, is binary:

- 1 = Customer Churn
- 0 = Customer Retention

C. Data Preprocessing

Prior to model development, the dataset underwent several preprocessing methods to improve the data's quality and ensure compatibility with our machine learning algorithms. Initially the dataset was inspected for missing values, duplicate records, and inconsistent data points. Duplicate observations were identified and removed to prevent bias during model training. The missing values were examined individually and appropriate processing techniques were used depending on the rest of the features.

Categorical features, such as Churn, were transformed into numerical representations so they can be analyzed by machine learning algorithms. Continuous numerical values were also examined for any possible anomalies or inconsistencies within the data, and re-evaluated to improve model convergence.

The dataset was then divided into independent predictor variables, X (PID, CRM PID Value, KA name, Active Subscribers, Not-Active Subscribers, Suspended Subscribers, Total Subscribers, Average Mobile Revenue, Average Fix Revenue, Total Revenue, ARPU), and the dependant target variable, y (Churn), representing customer status.

Finally the dataset was partitioned into training and testing subsets using the 80:20 split. The training data was used to develop the model and to do hyperparameter optimization, while the testing dataset remains hidden from the model during training, and is used only for the final performance evaluation.

D. Machine Learning Models

To determine the most effective artificial intelligence model for customer churn prediction, four primary machine learning algorithms were evaluated.

Logistic Regression

Logistic regression serves as a baseline for classification models. As a linear classifier, it estimates the probability the customer will churn based on historical data. Although simple compared to other neural network methods, Logistic Regression provides an insightful benchmark against more advanced models.

K-Nearest Neighbors(KNN)

The K-Nearest Neighbors algorithm classifies customers according to the results of neighboring customers with similar characteristics. For each customer in the testing dataset, the algorithm identifies the nearest data points within the training dataset and predicts churn based on the majority features among those instances. This approach uses customer similarity to guide prediction without requiring assumptions regarding feature relationships.

Random Forest

Random Forest is a machine learning algorithm that combines multiple decision trees to improve predictive accuracy. Each decision tree is trained using a randomly selected subset of observations and predictor variables within the training dataset. Final predictions are determined through majority voting across all trees within the forest. Because customer churn usually depends on complex nonlinear relationships among customer behaviors, Random Forest was expected to outperform simpler linear models.

Multi-Layer Perceptron(MLP)

The Multi-Layer Perceptron is a feedforward artificial neural network capable of learning complex nonlinear relationships between customer attributes and churn behavior. The network consists of interconnected neurons organized into hidden layers, allowing it to identify patterns that are not usually learned by traditional statistical models. Neural networks are really valuable for detecting subtle interactions among multiple customer characteristics that contribute to churn risk.

E. Feature Scaling

Many machine learning algorithms are sensitive to the vast changes in the scale of input features. Within the telecommunications dataset, numerical variables such as customer tenure, monthly charges, and service usage range substantially. Without normalization, variables with larger magnitudes can disproportionately influence the learning process, reducing model performance.

To address this issue, feature scaling was performed using the StandardScaler framework provided by the Scikit-learn library. The scaler was first fitted exclusively on the training dataset to calculate the mean and standard deviation for each numerical feature. These statistics were then used to transform both the training and testing dataset. By fitting the scaler only on the training data and applying the learned transformation to the testing data, we could make an unbiased evaluation of model performance.

Feature scaling is particularly important for optimization algorithms such as the Multi-Layer Perceptron (MLP) Classifier. Standardizing the input features improves numerical stability and helps prevent individual variables with larger numerical ranges from dominating the learning process.

The MLP model employed in this study used the Adam optimization algorithm with two hidden layers consisting of 100 and 50 neurons. The model was configured with a maximum of 500 training iterations and also used early stopping, allowing training to end automatically if validation performance failed to improve after 20 consecutive iterations.

F. Class Imbalance

Customer churn prediction datasets frequently have class imbalance, where the number of customers who remain with the telecommunications provider substantially exceeds the number of customers who discontinue service. If left unaddressed, this imbalance may incorporate bias into machine learning algorithms toward predicting the majority class. As a result, the model poorly identifies customers who are actually at risk of churning.

Before model development, the distribution of the target variable was examined within the training dataset to see how much imbalance existed. The analysis revealed 2,283 non-churn customers and 383 churn customers, highlighting that the minority class represented only a small proportion of the training data. This imbalance would have led to models achieving high overall accuracy while failing to correctly identify customers who are likely to churn.

To combat this challenge, we implemented the Synthetic Minority Oversampling Technique (SMOTE), from the imbalanced-learn library. This was applied exclusively to the training dataset after feature scaling. SMOTE generates synthetic examples of the minority class by interpolating between neighboring minority-class observations within the feature space rather than simply duplicating existing records. This structure preserves the underlying structure of the data while providing machine learning algorithms with a more balanced representation of both classes.

After oversampling, the training dataset contained 2,283 non-churn customers and 2,283 churn customers, an equal class distribution for model training. The testing dataset was intentionally left unchanged to preserve the original class distribution and ensure that model evaluation reflected realistic, proper customer populations that are encountered in real-world telecommunications environments.

G. Hyperparameter Optimization

Machine learning model performance is highly dependent upon the selection of appropriate hyperparameters. Therefore, hyperparameter tuning is used to improve predictive performance while minimizing overfitting. For the Random Forest classifier, multiple combinations of parameters, including the number of decision trees (n_estimators) and maximum tree depth (max_depth), were evaluated. The configuration that produced the strongest validation performance was selected for final model training.

H. Model Evaluation

To evaluate the effectiveness of each machine learning model, multiple performance metrics were used rather than relying only on overall classification accuracy. Because customer churn prediction is a binary classification problem, a combination of evaluation metrics can show a more comprehensive assessment of each model's predictive capabilities.

Confusion Matrix

The primary evaluation tool used in this study is the confusion matrix, which summarizes the performance of a binary classifier by comparing predicted customer classifications with the actual customer outcomes. We generated a confusion matrix for every machine learning model, allowing direct comparison of prediction behavior across algorithms.

The confusion matrix consists of four possible prediction outcomes (as shown in Figure 1).

	Predicted Churn	Predicted Non-Churn
Actual Churn	True Positive (TP)	False Negative (FN)
Actual Non-Churn	False Positive (FP)	True Negative (TN)

Figure 1: Structure of the confusion matrix

Each component of the confusion matrix provides a valuable insight:

- True Positive (TP): Customers correctly predicted to churn.

- True Negative (TN): Customers correctly predicted to remain with the telecommunications provider.
- False Positive (FP): Customers incorrectly predicted to churn when they actually remained.
- False Negative (FN): Customers incorrectly predicted to remain when they ultimately churned.

For each machine learning model developed in this study, including Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and the Multi-Layer Perceptron (MLP), a confusion matrix was generated using the testing dataset. These matrices provided a visual comparison of each algorithm's ability to correctly distinguish between churning and non-churning customers.

Accuracy

Accuracy measures the proportion of correctly classified observations among all customer records and serves as an overall indicator of model performance. Although accuracy provides a useful summary of predictive performance, it can be misleading when class imbalance exists because a model may achieve high accuracy simply by predicting the majority class.

Precision

Precision measures the proportion of customers predicted to churn who actually churned. A high precision value indicates that customer retention efforts are directed toward individuals who actually have a high risk of churn, therefore reducing unnecessary retention costs.

Recall

Recall, also known as sensitivity or the true positive rate, measures the proportion of actual churning customers correctly identified by the model. Within customer churn prediction, recall is particularly important because failing to identify customers who eventually leave may result in revenue loss.

F1-Score

The F1-score combines precision and recall into a single performance metric by calculating their mean. This metric is especially valuable for imbalanced classification problems because it balances both false positives and false negatives, indicated from the confusion matrix.

Analysis + Results

A. Model Performance Evaluation

To evaluate the effectiveness of different machine learning approaches for predicting customer churn, six classification models were developed and compared: Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, Multilayer Perceptron (MLP), a scaled MLP neural network, and an MLP model trained using Synthetic Minority Oversampling Technique (SMOTE). Because customer churn datasets are typically imbalanced, where non-churning customers significantly outnumber customers who leave, model performance was evaluated using multiple metrics including accuracy, precision, recall, and F1-score.

B. Logistic Regression Performance

The initial Logistic Regression model achieved an accuracy of 85.46%. However, despite achieving relatively high overall accuracy, the model demonstrated limited ability to identify customers likely to churn.

The classification report showed that the model achieved:

- Class 0 precision: 87%
- Class 0 recall: 97%
- Class 1 precision: 55%
- Class 1 recall: 18%
- Class 1 F1-score: 27%

The confusion matrix indicated that the model correctly classified 82 out of 100 churn customers, while incorrectly classifying 18 churn customers as non-churn customers. However, the model also produced a high number of missed churn predictions, demonstrating that Logistic Regression favored the majority class. (as shown in Figure 2)

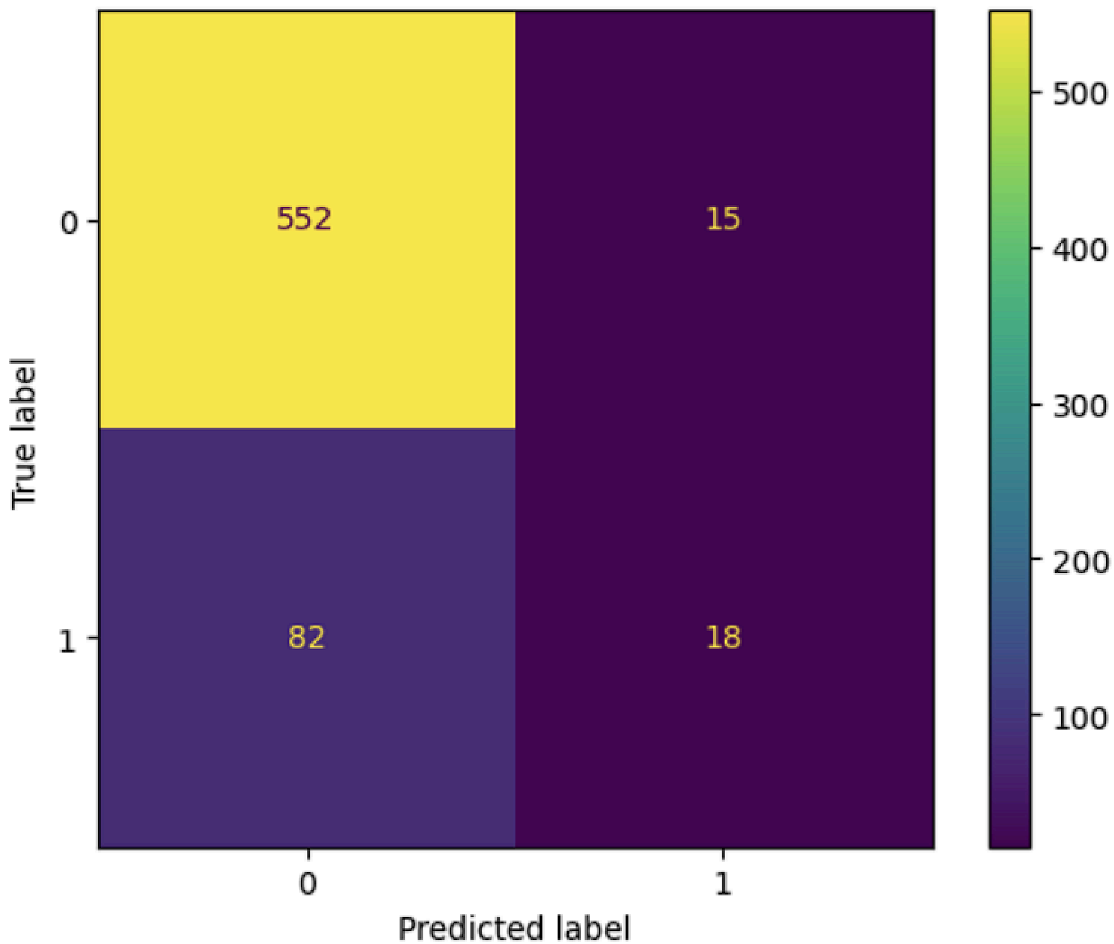

Figure 2: Logistic Regression confusion matrix

Although Logistic Regression provided a strong baseline, its low recall for churn customers suggests that it would not be sufficient for a telecommunications company attempting to proactively retain customers.

C. K-Nearest Neighbors Performance

The K-Nearest Neighbors model with three neighbors achieved an overall accuracy of 85.91%, slightly outperforming Logistic Regression. However, similar limitations remained regarding churn identification.

The model achieved:

- Class 0 precision: 89%
- Class 0 recall: 95%
- Class 1 precision: 55%
- Class 1 recall: 33%
- Class 1 F1-score: 41%

D. Random Forest Performance

The Random Forest classifier produced a substantial improvement compared to the previous models. Using a maximum depth of 80, the model achieved an accuracy of 96.40%.

The model achieved:

- Class 0 precision: 96%
- Class 0 recall: 99%
- Class 1 precision: 96%
- Class 1 recall: 79%
- Class 1 F1-score: 87%

Random Forest significantly improved churn prediction because decision trees can capture complex nonlinear relationships between customer characteristics. Unlike linear models, Random Forest can model interactions between variables such as customer tenure, monthly charges, contract type, and service usage patterns.

Hyperparameter testing was performed by analyzing the relationship between the number of estimators and model performance. Increasing the number of decision trees improved model stability, with performance plateauing after approximately 90 estimators. [Figure 3]

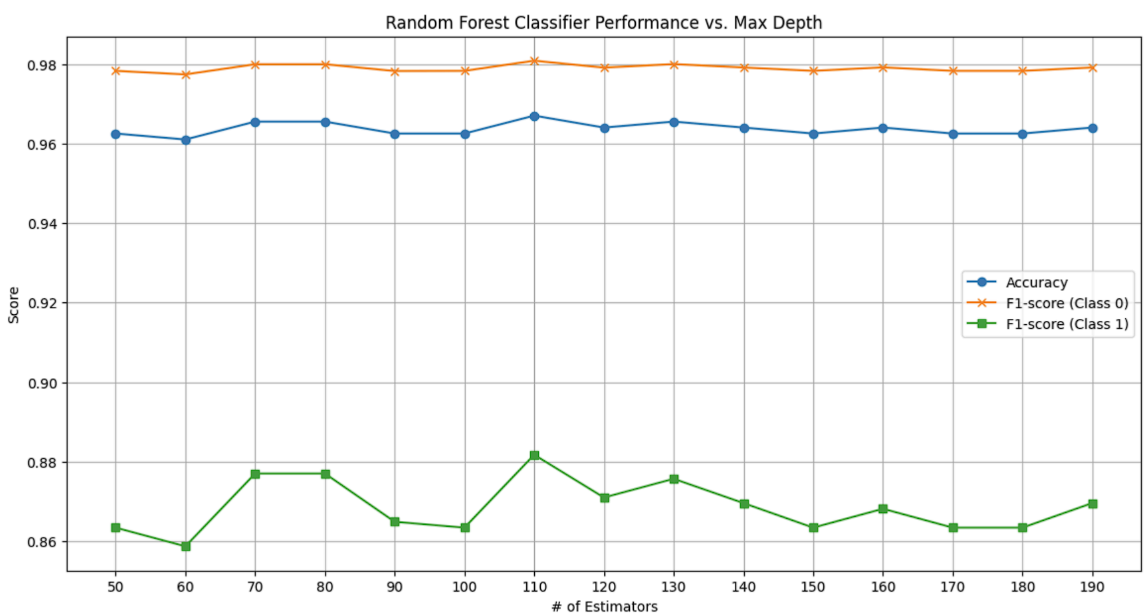

Figure 3: Hyperparameter tuning of the Random Forest classifier. Model performance plateaus after approximately 90 estimators, indicating diminishing returns from adding additional decision trees.

Additional testing was conducted using class weighting to address dataset imbalance. When using:

- max_depth = 20
- n_estimators = 90
- class_weight = "balanced"

the model produced a confusion matrix of [Figure 4]:

- True Negatives: 563
- False Positives: 4
- False Negatives: 20
- True Positives: 80

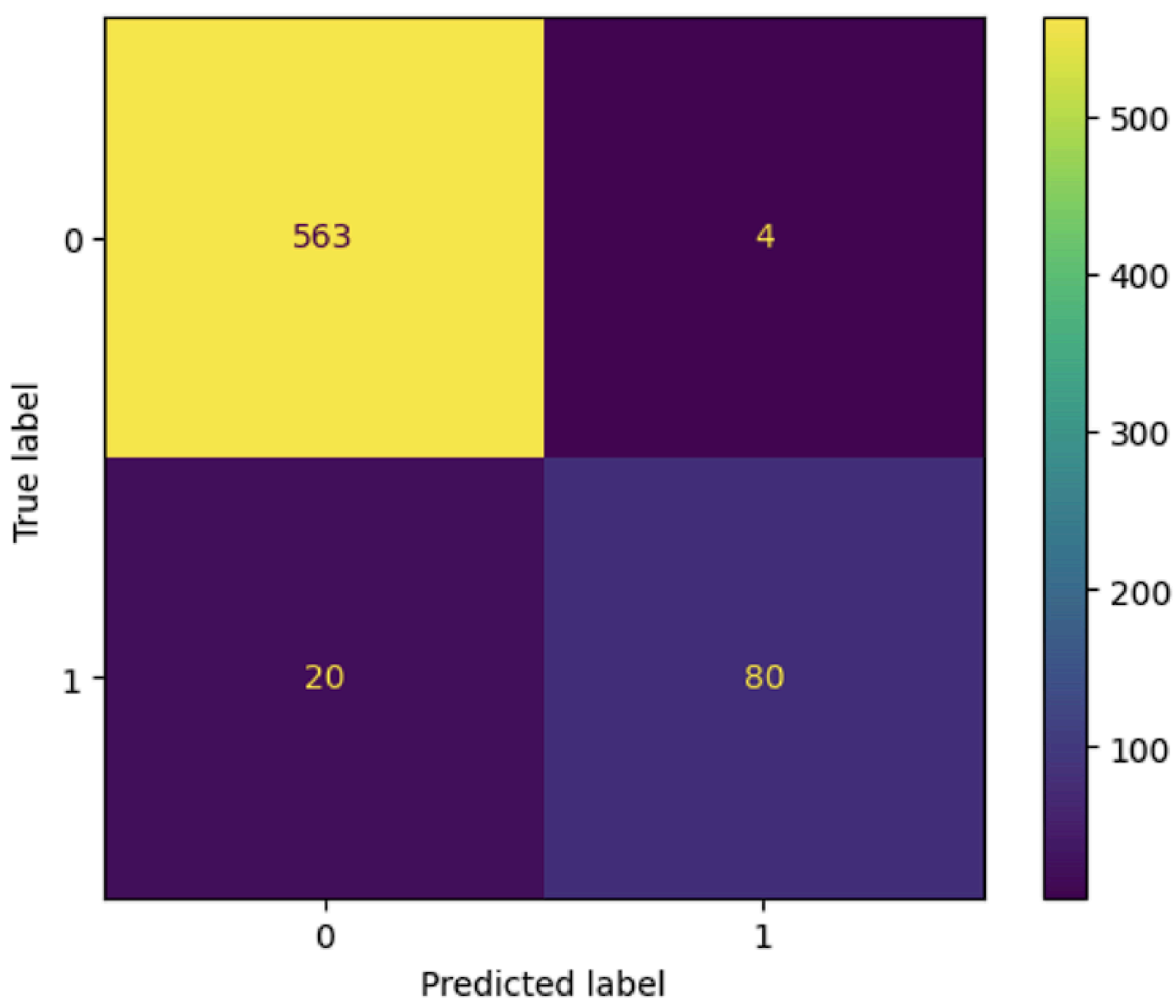

Figure 4: Confusion matrix *after* hyperparameter tuning

This version increased churn recall from the standard Random Forest model by prioritizing minority-class detection. The reduction in false negatives demonstrates the importance of adjusting models for imbalanced classification problems.

E. Neural Network Performance

A Multilayer Perceptron (MLP) neural network was initially tested without feature scaling. The model achieved an accuracy of 85.01%, but failed to identify churn customers.

Performance metrics included:

- Class 1 precision: 0%
- Class 1 recall: 0%
- Class 1 F1-score: 0%

The model classified nearly all customers as non-churn, demonstrating that neural networks require proper preprocessing and sufficient optimization before producing meaningful results.

F. Scaled Neural Network Performance

After applying StandardScaler normalization and increasing the network complexity to two hidden layers (100 and 50 neurons), the MLP model achieved significantly improved performance.

The scaled MLP achieved:

- Accuracy: 94.30%
- Class 0 recall: 98%
- Class 1 precision: 85%
- Class 1 recall: 75%
- Class 1 F1-score: 80%

The confusion matrix showed [Figure 5]:

- True Negatives: 554
- False Positives: 13
- False Negatives: 25
- True Positives: 75

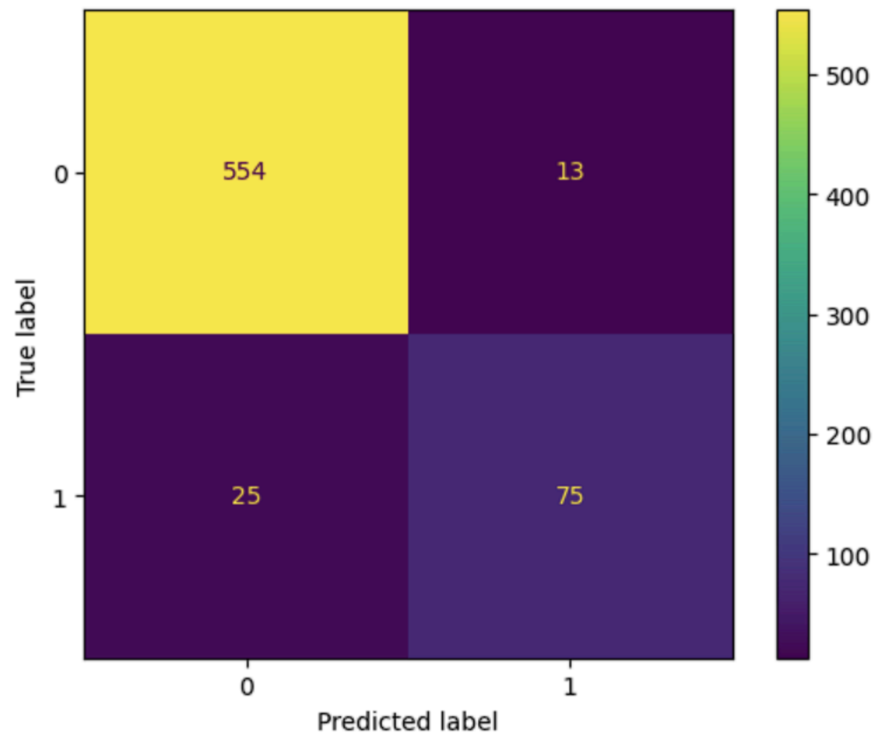

Figure 5: Confusion matrix of the MLP model *after* applying StandardScaler normalization

Feature scaling greatly improved neural network performance because MLP models rely on gradient-based optimization, which performs more effectively when numerical features exist on comparable scales.

Compared with Random Forest, the MLP achieved slightly lower accuracy but demonstrated competitive churn detection ability.

G. SMOTE-Based Neural Network Performance

Because the original dataset contained significantly fewer churn customers, SMOTE was applied to balance the training dataset. Before oversampling, the training data contained:

- Non-churn customers: 2,283
- Churn customers: 383

After applying SMOTE, both classes contained 2,283 samples.

The SMOTE-based MLP achieved:

- Accuracy: 89.81%
- Class 1 precision: 65%
- Class 1 recall: 69%

- Class 1 F1-score: 67%

The confusion matrix showed [Figure 6]:

- True Negatives: 530
- False Positives: 37
- False Negatives: 31
- True Positives: 69

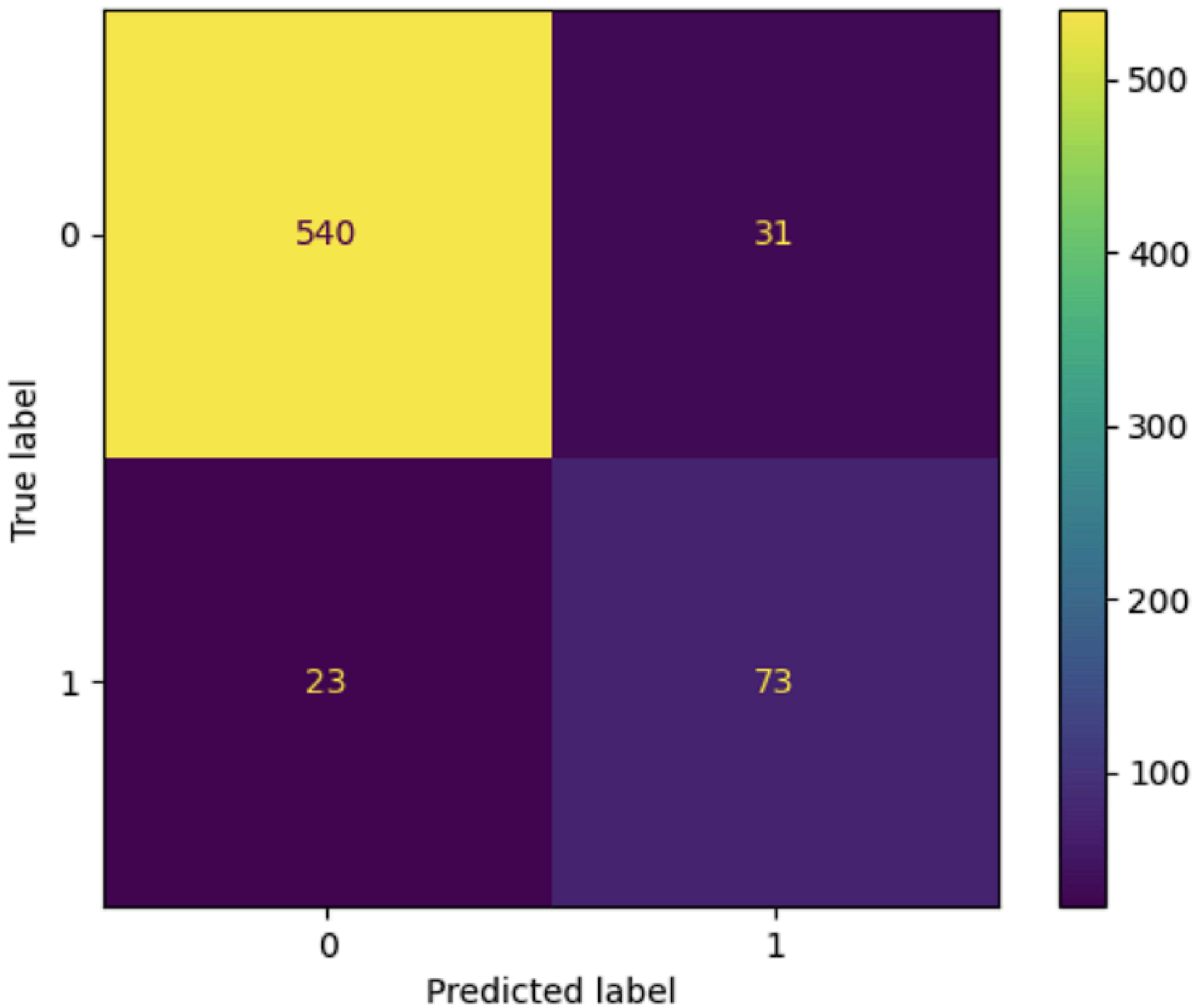

Figure 6: Confusion matrix of the MLP model *after* SMOTE

Although SMOTE reduced overall accuracy compared to the original scaled MLP, it improved the model's ability to recognize churn customers compared to models trained on imbalanced data. This demonstrates the tradeoff between maximizing general accuracy and improving minority-class detection.

H. Analysis

Across all tested models, Random Forest produced the strongest overall predictive performance, achieving the highest accuracy (96.40%) and the strongest churn classification metrics, with a Class 1 F1-score of 87%. This indicates that Random Forest provided the best balance between identifying churn customers and avoiding unnecessary retention interventions.

The scaled MLP model achieved the second strongest performance, with a Class 1 F1-score of 80%, demonstrating that neural networks can effectively model churn behavior when appropriate preprocessing is applied.

Traditional models such as Logistic Regression and KNN achieved acceptable accuracy but struggled with churn identification. Their low recall values indicate that they frequently classified actual churn customers as loyal customers, which would limit their usefulness in a business environment.

The SMOTE-based MLP model demonstrated that addressing class imbalance improves minority-class detection. While it decreased overall accuracy, it increased the model's ability to identify churn customers by reducing false negatives.

Overall, the results demonstrate that machine learning models can successfully predict customer churn with high accuracy. The findings suggest that ensemble learning methods, particularly Random Forest, provide the strongest foundation for developing proactive customer retention systems. By identifying customers at high risk of leaving before churn occurs, telecommunications companies can implement targeted interventions, potentially reducing customer loss and increasing long-term revenue.

Discussion

The analysis of this study demonstrates that artificial intelligence and machine learning can successfully predict customer churn in the telecommunications industry. Although traditional machine learning models, such as Logistic Regression and K-Nearest Neighbors achieved an overall solid score, they struggled to correctly identify real customers who were at risk of churning. Due to the dataset being imbalanced, with non-churning customers outweighing actual churning customers, these models favored the majority class. In a real world business scenario this limitation is consequential as failing to identify the real customers who are at risk of leaving represents missed opportunities for targeted customer retention efforts.

Conversely, the more complex models, specifically Random Forest and the scaled Multilayer Perceptron, showed significantly stronger performance. The Random Forest classifier achieved the highest overall accuracy of 96.4% while maintaining a Class 1 F1-score of 87%, indicating that it effectively balanced precision and recall when identifying churning customers. The scaled MLP model also performed adeptly after feature normalization, showing the importance of proper data preprocessing for neural networks. Without feature scaling, the neural network had failed to detect official churn customers. On the other hand, scaling allowed the model to learn meaningful relationships among customer attributes and significantly improved binary classification performance.

Moreover, the application of SMOTE revealed the challenges with class imbalance. Oversampling the minority class reduced overall accuracy by a margin, but improved the model's ability to correctly identify churn customers more consistently. This tradeoff represents a common challenge in real-world machine learning applications. A company may accept a small reduction in overall accuracy if it means identifying more customers who are genuinely at risk of leaving. The cost of offering a retention incentive to a loyal customer is usually lower than the revenue lost when an existing customer permanently switches providers.

From a business point of view, these results suggest that predictive artificial intelligence systems can become a valuable decision-support tool rather than a replacement for human decision-making. Instead of contacting every customer with retention offers, companies can prioritize outreach toward a certain group of customers identified as high-risk by our predictive model. This targeted approach has the potential to reduce unnecessary expenditures while increasing customer retention rates in telecommunications businesses.

Despite these promising results, several limitations can be identified. First, this study used a single, foreign and publicly available telecommunications dataset, which may not fully represent customer behavior across different providers or geographic regions, especially in the United States. Customer preferences, pricing structures, and service offerings vary between companies, potentially affecting model performance when applied to new datasets.

Second, the dataset primarily consisted of structured customer information such as contract type, monthly charges, tenure, and service subscriptions. Current and more modern telecommunications companies often contain significantly richer sources of information, including customer support interactions, network performance metrics, billing history, behavioral usage patterns, etc. Incorporating these additional variables can further improve predictive performance.

Finally, this research solely focused on predicting whether a customer would churn in the near future rather than estimating *when* exactly the churn would occur or evaluating the effectiveness of a specific retention strategy to keep a customer. Although prediction is an important first step, the next phase would need to determine optional interventions for a specific customer to maximize long-term profitability.

Future research can build upon the findings in this study through many different approaches. More advanced machine learning techniques such as XGBoost, LightGBM, and CatBoost could be evaluated and compared against our current Random Forest and feature-scaled MLP model. Additionally, modern machine learning architectures capable of modeling sequential customer behavior, including recurrent neural networks (RNNs) and transformer-based models, may also improve predictive performance when more data is available. Finally, future studies could integrate explainable AI techniques such as SHAP or LIME to identify which customer characteristics contribute most strongly to churn predictions. Improving model interpretability would definitely increase confidence among business owners and investors and support more structured retention strategies.

Conclusion

Customer churn continues to be one of the most significant challenges for companies, mainly telecommunications businesses, because acquiring new customers is substantially more expensive than retaining existing ones. Therefore, developing accurate predictive models has the potential to generate meaningful financial benefits by enabling companies to identify high-risk customers before they discontinue their service.

This study researched whether artificial intelligence and machine learning models could accurately predict customer churn using historical customer data. Six machine learning approaches were evaluated, including Logistic Regression, K-Nearest Neighbors, Random Forest, Multilayer Perceptron, a scaled neural network, and a SMOTE-enhanced neural network. Performance was assessed using accuracy, precision, recall, F1-score, and a confusion matrix analysis.

Across all the evaluated models, the Random Forest classifier demonstrated the strongest overall performance, achieving an accuracy of 96.4% while maintaining excellent predictive capability for churn customers. The scaled neural network also produced strong results, demonstrating the importance of feature preprocessing for advanced machine learning models. Models trained on imbalanced data generally favored the majority class, while techniques such as SMOTE improved minority-class detection despite reducing overall accuracy.

In the end, these findings indicate that artificial intelligence can accurately predict customer churn with a high degree of accuracy and, consequently, has the potential to support more effective customer retention strategies within companies. Rather than relying only on reactive approaches after customers decide to leave, companies can leverage predictive analytics to identify current customers at risk early on and implement targeted plans. Such proactive strategies could possibly improve customer satisfaction, reduce churn-related revenue and profit losses, and, finally, allocate retention resources more efficiently.

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